Architecture-dependent effects of imbalance correction in melanoma classification: a controlled audit with fairness, calibration, and explanation analysis

Nursultan Zhanat[1,2] and Sun Baoshan[1,2,*]

[1] School of Computer Science and Technology, Tiangong University, Tianjin 300387, China
[2] Tianjin Key Laboratory of Autonomous Intelligence Technology and Systems, Tiangong University, Tianjin 300387, China

[*] Corresponding author: Sun Baoshan, sunbaoshan@tiangong.edu.cn, ORCID 0000-0003-2193-3232
Nursultan Zhanat: zhanat.nursultan1@gmail.com, ORCID 0009-0006-7839-4785

Statements and Declarations

Funding

This work received no specific funding.

Competing interests

The authors declare no competing interests.

Author contributions

N.Z. designed the study, implemented the code, ran the experiments, and wrote the manuscript. S.B. supervised the project and revised the manuscript. Both authors have read and agreed to the published version of the manuscript.

Ethics approval

Not applicable. All data are public, de-identified research datasets, and the study involved no patient contact or new data collection.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Data availability

HAM10000 is available from the ISIC archive. HIBA is available from the ISIC archive under a CC-BY license. The per-sample test predictions, the lesion-aware split file, the contrast features, fairness summaries, XAI metrics, and external-validation results are released with the code repository.

Code availability

The code is at `https://github.com/conorkz/melanoma-xai-bias`. The repository contains the training configs, split construction, evaluation scripts, table generators, and tests. The submission version is tagged `v1.0-submission`.

Acknowledgements

The authors thank the maintainers of the public HAM10000 and HIBA datasets.